SAME SIZE ARTWORK
LEAFLET SIZE: 140 mm x 88 mm

20 mm

22 mm

22 mm

to 10 days. The area may be covered with a dressing or occluded if desired.

Route of administration : Topical

SYMPTOMS AND TREATMENT FOR OVERDOSAGE AND ANTIDOTE (S)

No overdosage has been reported with mupirocin ointment. Since systemic absorption of mupirocin ointment is very low, overdosage is less likely with mupirocin ointment.

PRESENTATION

5g and 15 g tube

SHELF LIFE

18 months

STORAGE CONDITIONS, USER INSTRUCTIONS AND PHARMACEUTICAL PRECAUTIONS

Store below 30°C. Do not freeze.

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BRU20032821P

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Product Registration Holder
Glenmark Pharmaceuticals (Malaysia)
Sdn Bhd (660397-W)
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Manufactured by:
glenmark
PHARMACEUTICALS LTD.
Plot No. E-37, 39, D-Road,
MIDC, Satpur, Nashik - 422 007,
Maharashtra State, India.
TM Trade Mark

PE00000 **MY**

For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory.

Supirocin™
Ointment

Mupirocin Ointment USP

SPACE FOR
PHARMACODE

COMPOSITION:

Mupirocin USP	2% w/w
Ointment base	q.s.

DESCRIPTION

A printed carton containing an insert and a printed aluminum collapsible tube containing a white semi-solid ointment.

THERAPEUTIC CLASS

D06AX09

PHARMACOLOGY

Mechanism of action

Mupirocin (pseudomonic acid; pseudomonic acid A) is an antibiotic produced by submerged fermentation of *Pseudomonas fluorescens* and is structurally unrelated to other antibiotics. Other substances produced by *Pseudomonas fluorescens* which are being investigated for antibacterial activity include pseudomonic acids B, C, and D. However, mupirocin appears to have the most antibacterial activity. Pseudomonic acids B, C, and D are generally 2 to 4-fold less active than pseudomonic acid A (mupirocin).

The mechanism of action of mupirocin is unique in that it reversibly and specifically binds to bacterial isoleucyl transfer-RNA synthetase resulting in arrest of protein synthesis, a mechanism dissimilar from any other antibiotic, and inhibiting RNA synthesis. This mechanism of action presumably accounts for the slow rate of emergence of mupirocin-

SAME SIZE ARTWORK
LEAFLET SIZE: 140 mm x 88 mm

resistant staphylococci observed in *in vitro* studies and lack of cross-resistance with other topical antibiotics. The drug has been active against multi-resistant strains of *Staphylococcus aureus* and coagulase-negative *staphylococci*, including strains resistant to penicillin, methicillin, neomycin, erythromycin, fusidic acid, lincomycin, chloramphenicol, tetracycline and gentamicin.

Mupirocin enters cells by passive diffusion. The concentration of mupirocin in sensitive bacteria is dependent upon binding sites on isoleucyl-tRNA synthetase. Resistance to mupirocin involve restricted access to the binding site on isoleucyl-tRNA synthetase as opposed to changes in the cell membrane.

Mupirocin is used for the topical treatment of impetigo due to Staphylococcus aureus; Group A beta-hemolytic *Streptococcus*, and *Streptococcus pyogenes*.

Antimicrobial spectrum

Based on *in vitro* studies, mupirocin has a broad range of antibacterial activity. Gram-positive bacteria sensitive to mupirocin include *Staphylococcus aureus* (including methicillin-resistant and beta-lactamase producing), Staphylococcus epidermis, Staphylococcus saprophyticus, and Streptococcus pyogenes. Gram-negative bacteria sensitive to mupirocin include H influenzae, N gonorrhoeae, N meningitidis, Bordetella pertussis, Pasteurella multocida and M catarrhalis (Minimum inhibitory concentration (MIC) less than or equal to 0.25 mg/L). The drug has also been active *in vitro* against methicillin-resistant S. aureus that is resistant to gentamicin.

Mupirocin is bactericidal at concentrations achieved by topical administration. However, the minimum bactericidal concentration (MBC) against relevant pathogens is generally eight-fold to thirty-fold higher than the minimum inhibitory concentration (MIC). In addition, mupirocin is highly protein bound (>97%), and the effect of wound secretions on the MICs of mupirocin has not been determined.

Resistance occurs rarely. However, when mupirocin resistance does occur, it appears to result from the production of a modified isoleucyl-tRNA synthetase. High-level plasmid-mediated resistance (MIC >1024 mcg/mL) has been reported in some strains of S. aureus and coagulase-negative staphylococci. Due to this unique mode of action, mupirocin demonstrates no *in vitro* cross-resistance with other classes of antimicrobial agents.

Pharmacokinetics

The bioavailability of a topical mupirocin formulation is very low (0.24%). Penetration of mupirocin is enhanced if occlusive dressing is used or if mupirocin is applied to damaged or diseased skin. Measurable radioactivity was present in the stratum corneum of these subjects 72 hours after application. Mupirocin is slowly metabolized in the skin to an inactive metabolite, monic acid. Only 2.7% of mupirocin is metabolized in 48 hours in homogenates of human skin.

INDICATIONS

Supirocin Ointment is indicated for the bacterial skin infections e.g: impetigo, folliculitis and furunculitis.

CONTRAINDICATIONS

Hypersensitivity to mupirocin or to any components in the formulation.

SIDE EFFECTS / ADVERSE REACTIONS

Adverse reactions are listed below by system organ class and frequency. Frequencies are defined as: very common (1/10), common (1/100, <1/10), uncommon (1/1000, <1/100), rare (1/10,000, <1/1000), very rare (<1/10,000), including isolated reports. Very rare adverse reactions were primarily determined from post-marketing experience data and therefore refer to reporting rate rather than true frequency.

Skin and subcutaneous tissue disorders:

Common: Burning localised to the area of application.
Uncommon: Itching, erythema, stinging and dryness localised to the area of application.
Uncommon: Cutaneous sensitisation reactions to mupirocin or the ointment base.

Immune system disorders:

Very rare: Systemic allergic reactions have been reported with Mupirocin Ointment.

PRECAUTIONS / WARNINGS

- Formulations containing polyethylene glycol should not be applied to broken skin or mucus membranes, extensive open wounds or to burns.
- Formulations containing polyethylene glycol should not be used in patients with moderate or severe renal impairment; absorption of polyethylene glycol is possible.
- Avoid contact with eyes.
- As with other antibacterial products, prolonged use may result in overgrowth of non susceptible organisms, including fungi.

DRUG INTERACTION

The effect of the concurrent application of Mupirocin ointment and other drug products has not been studied.

USE IN PREGNANCY AND LACTATION

Pregnancy

Pregnancy Category B

Animal studies using doses that exceed the recommended human dose demonstrated that mupirocin is not teratogenic. However, there are no well-controlled studies in pregnant women. Animal teratogenicity studies are not always predictive of human response. Mupirocin should be used during pregnancy only if the drug is clearly needed.

Lactation

It is unknown if mupirocin is excreted in human milk; caution should be used when mupirocin is administered to a nursing woman.

RECOMMENDED DOSAGE, DOSAGE SCHEDULE AND ROUTE OF ADMINISTRATION

Adults and children:

Supirocin Ointment should be applied to the affected area up to three times daily, for up

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ICONGRAPHICS CODE:

PANTONE SHADE

PANTONE BLACK PROCESS C

Supersedes Artwork Code PE48924

PHARMACODE: